

UTAG: Leveraging LLM as a Unified Embedding Generator for Text-Attributed Graphs

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Abstract

Large Language Models (LLMs) have shown significant potential in handling text-attributed graphs (TAGs). Nevertheless, existing LLM-enhanced approaches for TAG learning mainly utilize LLMs as text augmenters, which not only incur substantial computational overhead but may also introduce noise. In contrast, their capacity as embedding generators remains underutilized. Furthermore, most current methods are designed for specific TAG instances and lack cross-domain transferability. To overcome these limitations, we propose **UTAG**, a novel framework that leverages LLMs as unified embedding generators. First, we design a dual-view contrastive fine-tuning approach for cross-domain pretraining. The in-domain view promotes local consistency via adaptive positive sample selection, while the cross-domain view establishes alignment through subspace clustering-based cross-domain prototype aggregation. By jointly pretraining on multiple TAG datasets from diverse domains, we obtain a unified LLM encoder that produces transferable representations. Subsequently, we introduce a graph-specific adaptation training technique that implements two-stage dimensionality reduction with carefully designed semantic and Laplacian reconstruction constraints. This allows effective incorporation of structural signals while preserving semantic information to the greatest extent. Additionally, we employ cross-space alignment to further enhance consistency. Extensive experiments demonstrate that UTAG substantially surpasses existing LLM-enhanced methods, achieving an average improvement of 5.9% on cross-domain transfer tasks, which verifies its effectiveness and superiority.

CCS Concepts

• Information systems → Data mining.

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Keywords

Text-Attributed Graph; Large Language Model; Graph Neural Network; Cross-domain Transfer

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1 Introduction

Text-attributed graphs (TAGs) [14, 21] represent an important class of graph-structured data prevalent in real-world scenarios like academic citation networks [17], social media, and knowledge graphs [33]. In TAGs, nodes possess both textual attributes and structural relationships, which collectively reflect their intrinsic properties. Node representations directly influences the performance of downstream tasks such as node classification and link prediction.

Early approaches to TAG representation learning typically encoded textual content using shallow features [37], which were then fed into graph neural networks (GNNs). Although computationally efficient and scalable, these methods often failed to capture fine-grained semantic information. With the advent of pre-trained language models (LMs) [24], a growing body of research has sought to better integrate LM-generated text representations with graph structures. Though significantly improving semantic expressiveness, these approaches still face substantial training overhead and insufficient structure-semantics alignment.

Recently, Large Language Models (LLMs) [3, 28] have been increasingly applied to graph-related tasks [5, 32], demonstrating promising performance in TAG representation learning. For instance, TAPE [14] employs LLMs to generate explanatory texts, which are subsequently translated into features for downstream tasks. GAUGLLM [10] introduces a mixture-of-prompt-experts approach, utilizing three distinct prompt templates to generate augmented node descriptions. LLM4NG [36] leverages LLMs to extract semantic information from labels and generate synthetic samples.

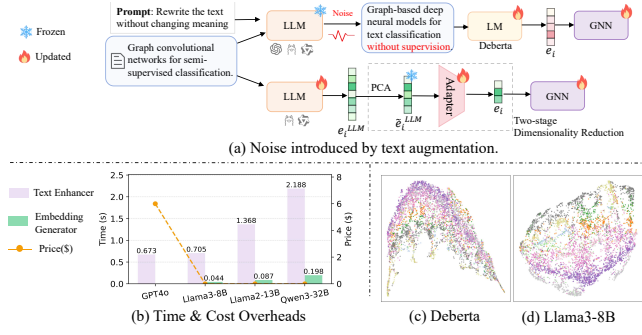


Figure 1: A Comparative Analysis of LLM Utilization Paradigms: Text Augmentation vs. Embedding Generation. (a) Architecture Comparison and Noise Characterization, (b) Analysis of Computational and Monetary Costs, (c) and (d) Latent Space Distributions of LM and LLM Embeddings.

However, most existing LLM-enhanced methods primarily treat LLMs as text augmenters [21, 36], and employ LMs to produce embeddings. While these approaches have shown effectiveness, they often entail high computational costs [4] and risk introducing noise [8], as illustrated in Figure 1(a) and (b). Furthermore, embeddings generated by language models such as DeBERTa exhibit significant cluster overlap compared to those from more advanced LLMs, as visualized in Figure 1(c) and (d). These limitations highlight the under-explored potential of LLMs as direct embedding generators. In addition, most existing techniques are tailored to specific TAG datasets and lack cross-domain transferability [41]. To address these limitations, we investigate using LLMs as unified embedding generators applicable to multiple datasets across diverse domains. Unlike unified graph backbones such as UniGraph [15] and UniGLM [9] that focus on architectural unification, our goal is to achieve representation-level unification by aligning semantic information within a shared embedding space.

Nevertheless, directly employing LLMs as unified embedding generators still faces two major challenges: **(1) Lack of fine-grained semantic discrimination and cross-domain transferability.** Although general-purpose LLMs can encode TAG texts from different domains, their representations tend to capture global, coarse-grained semantics while struggling with fine-grained distinctions, limiting within-domain discriminability. Furthermore, distributional differences across TAGs cause domain shift, where embeddings effective in one domain lose expressiveness when transferred to another. **(2) Dimensional mismatch between high-dimensional semantic space and low-dimensional structural space.** LLMs typically produce embeddings with thousands of dimensions, whereas GNNs operate in low-dimensional structural spaces. Directly mapping LLM embeddings into the structural space leads to significant information loss [1]. This issue is compounded by the fact that the semantic and structural spaces are designed with divergent modeling objectives. Consequently, without carefully crafted alignment and constraint mechanisms, their integration is prone to imbalance.

To address these challenges, we propose **UTAG**, a novel framework that leverages LLMs as embedding generators through cross-domain pretraining and graph adaptation. During pretraining, we

design a dual-view contrastive fine-tuning method where the in-domain view enhances local consistency via adaptive positive sample selection with soft-assignment aggregation, and the cross-domain view achieves alignment through subspace clustering-based prototype aggregation. By jointly pretraining on multiple TAG datasets from diverse domains with a tailored contrastive loss, we align node texts into a shared semantic space, yielding a unified LLM encoder. In the adaptation stage, each TAG undergoes lightweight structural injection based on this shared encoder, enabling domain-specific adaptation while preserving semantic consistency. Specifically, we introduce a trainable adapter for two-stage dimensionality reduction that projects LLM embeddings into the graph-structural space. This is complemented by structural and semantic reconstruction constraints alongside cross-space alignment, ensuring node representations are both structure-aware and semantically informative. Our main contributions can be summarized as follows:

- We explore a novel perspective on TAG representation learning by leveraging LLMs as unified embedding generators, advancing the development of graph foundation models.
- We design a dual-view contrastive fine-tuning approach for cross-domain pretraining, producing a unified LLM encoder that effectively addresses cross-domain transfer challenges in TAGs.
- We propose a graph adaptation strategy with two-stage dimensionality reduction and structural-semantic constraints, effectively resolving the dimensional mismatch problem.
- Extensive experiments demonstrate that UTAG significantly outperforms existing LLM-enhanced approaches, achieving an average improvement of 5.9% in cross-domain transfer tasks, validating its effectiveness and superiority.

2 Related Work

2.1 LLM-Enhanced Methods for TAGs

A growing body of research has explored the use of LLMs to enhance performance on TAGs [10, 38]. Most existing approaches treat LLMs as textual augmenters, generating supplementary textual information based on original node attributes [35, 36]. For instance, TAPE [14] employs LLMs to create explanatory text for individual nodes, thereby enriching their contextual information. Similarly, LanSAGNN [21] utilizes LLMs to extract customized semantic features for node pairs. Although these methods have shown effectiveness, they heavily depend on the quality of LLM-generated content, which can be noisy. Furthermore, processing every node's text through an LLM incurs substantial computational and financial costs, particularly for large-scale graphs. In contrast, the paradigm of leveraging LLMs directly as embedding generators remains largely unexplored. A few studies, such as OFA [23], have introduced graph prompting paradigms for in-context learning. OFA uses an LLM to encode raw text and then applies a simple Multi-Layer Perceptron (MLP) to project high-dimensional embeddings into the low-dimensional space. However, this straightforward mapping approach inevitably leads to significant information loss [1].

2.2 Graph Foundation Models

Graph Foundation Models (GFM) aim to build versatile models for handling diverse graphs and tasks. Current GFM pre-training follows two main paradigms: language modeling pre-training [26] and

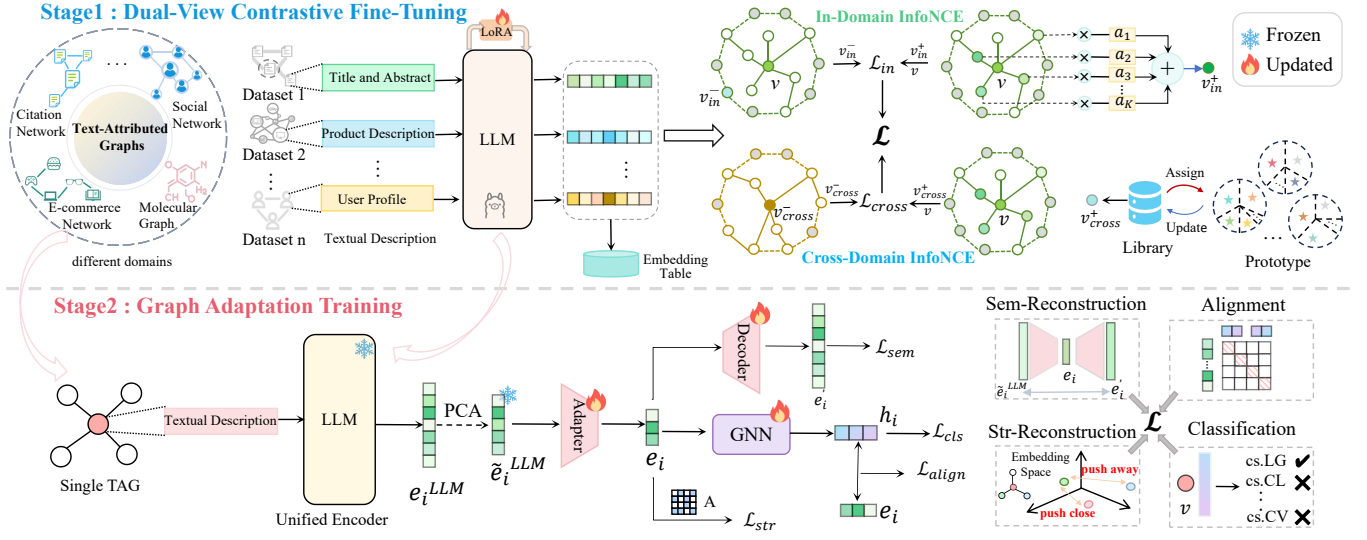


Figure 2: Overview of the proposed UTAG.

graph pre-training [27]. The language modeling approach adapts self-supervised pretext tasks from language to graphs. While pure LLMs show preliminary graph reasoning ability, the modality gap between text and graphs hinders performance without architectural support [11]. Studies like UniGraph [15] and P2TAG [39] integrate GNNs into LLM frameworks and jointly pre-train them using Masked Language Modeling (MLM). The graph pre-training paradigm, in contrast, adapts graph self-supervised tasks to pre-train or assist language model architectures. For instance, GraphCLIP [41] aligns graph-summary pairs via contrastive and invariant learning. Despite progress, most GFMs still rely on conventional LMs as backbones, with more powerful LLMs typically serving auxiliary roles rather than being central architectural components.

3 Preliminaries

Notations We consider a collection of text-attributed graphs (TAGs) from D domains, denoted as $\mathcal{G}^d = (\mathcal{V}^d, \mathcal{E}^d, \mathcal{T}^d)$ for $d = 1, 2, \dots, D$, where $\mathcal{V}^d$ is the node set, $\mathcal{E}^d$ is the edge set represented by adjacency matrix A^d , and each node $v_i^d \in \mathcal{V}^d$ is associated with a textual description $t_i^d \in \mathcal{T}^d$ (e.g., titles or reviews).

Problem Definition Given multiple TAG datasets $\{\mathcal{G}^d\}_{d=1}^D$ from different domains, we aim to learn a **unified encoder** f_θ that constructs a consistent semantic space for nodes across domains while maintaining cross-domain transferability. This encoder provides unified initial representations for subsequent structural modeling. Formally, the encoder is defined as:

$$f_\theta : \bigcup_{d=1}^D \mathcal{T}^d \rightarrow \mathbb{R}^{d_t}, \quad (1)$$

which generates embeddings for all nodes across domains as: $z_i^d = f_\theta(t_i^d)$, $\forall d \in \{1, \dots, D\}$, $\forall i \in \{1, \dots, n_d\}$. Based on these unified embeddings z_i^d , a second-stage graph adapter subsequently incorporates structural information from A^d to produce the final joint representations h_i^d for downstream tasks.

4 Method

The overall architecture of UTAG, shown in Figure 2, consists of two stages: (1) jointly pretraining a unified encoder through dual-view contrastive fine-tuning on multiple cross-domain TAG datasets for aligning semantic space, and (2) graph adaptation training that maps LLM embeddings generated by the unified encoder into the graph-structural space to achieve structural integration.

4.1 Dual-View Contrastive Fine-Tuning (DCFT)

Given multi-domain TAGs $\mathcal{G}^d$, this stage aims to train a unified text encoder f_θ that establishes a shared semantic representation space across domains. For any node i with text t_i^d in domain d , its semantic representation is encoded as $z_i^d = f_\theta(t_i^d)$ (abbreviated as z_i hereafter). We introduce two complementary views under a contrastive learning objective: the **in-domain view** and the **cross-domain view**. Both views are jointly optimized in the same embedding space using a tailored contrastive loss. To reduce optimization complexity and computational cost, we fine-tune the LLM using the LoRA [16] technique, updating only a small subset of parameters.

In-Domain View. To maintain consistency among node representations within local neighborhoods and capture fine-grained semantic distinctions, we propose the in-domain view. The core idea involves adaptive sample selection and weighted aggregation to generate local representative vectors, coupled with a contrastive objective to reinforce local consistency. Specifically, for an anchor node i , we construct a candidate set C_i comprising semantic neighbors retrieved via in-domain Top- K similarity search and structural neighbors from 1-hop and 2-hop graph connections. To identify nodes that preserve both semantic and structural similarity, we define a hybrid similarity score between node i and candidate j as:

$$s_{ij} = \frac{1 + \text{sim}(z_i, z_j)}{2} + \lambda_g (\alpha_1 \text{Jac}(i, j) + \alpha_2 \text{PPR}(i, j)) + \lambda_p \frac{|P_i \cap P_j|}{M}, \quad (2)$$

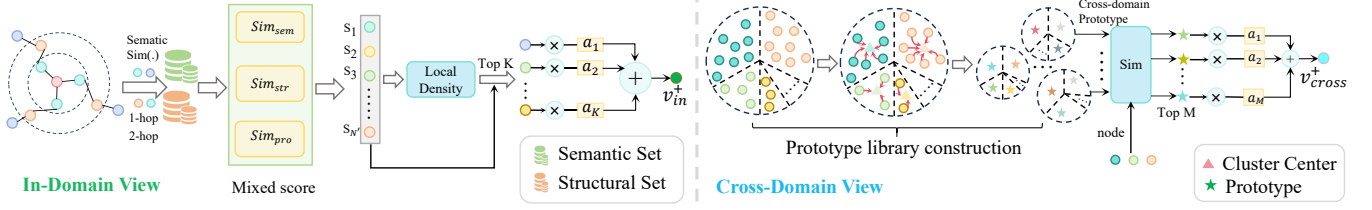


Figure 3: The details of In-Domain View and Cross-Domain View.

where $\text{Jac}()$ denotes Jaccard similarity and $\text{PPR}()$ represents normalized Personalized PageRank, capturing local and global structural similarity, respectively. Here, P_i denotes the Top- k most similar in-domain prototypes for node i , which will be introduced in the cross-domain view section.

To address varying graph region densities and prevent under-aggregation in dense regions or excessive noise in sparse regions, we introduce a density-adaptive sample selection mechanism. We first compute the local density of node i as:

$$\rho_i = \frac{1}{|C_i|} \sum_{j \in C_i} \max(s_{ij}, 0), \quad (3)$$

and obtain $\rho_{\min}$ and $\rho_{\max}$ using batch quantiles and exponential moving averages. The density is then mapped to an adaptive number of positive samples within the range $[K_{\min}, K_{\max}]$:

$$K_i = K_{\min} + (K_{\max} - K_{\min}) \cdot \frac{\rho_i - \rho_{\min}}{\rho_{\max} - \rho_{\min}}, \quad (4)$$

enabling nodes in dense regions to aggregate more positives for consistency, while nodes in sparse regions select fewer but more reliable neighbors to avoid noise.

We rank the candidate set by s_{ij} , select the top- K_i nodes to form the positive set S_i , and aggregate them using softmax weights with an adaptive temperature to obtain the local representative vector:

$$\tilde{z}_i = \text{norm}\left(\sum_{j \in S_i} \alpha_{ij} z_j\right), \text{ with } \alpha_{ij} = \frac{\exp(\gamma_i s_{ij})}{\sum_{u \in S_i} \exp(\gamma_i s_{iu})} \quad (5)$$

Here, the temperature coefficient γ_i is adaptively adjusted based on the variance of candidate scores, ensuring balance between dense and sparse regions. This weighted aggregation allows each node to incorporate diverse semantic and structural information from its neighborhood, resulting in a more robust and discriminative local representation $\tilde{z}_i$.

Finally, to enforce consistency between node i and its local representative, we adopt an InfoNCE-based contrastive objective:

$$\mathcal{L}_{in} = -\frac{1}{B} \sum_{i=1}^B \left(\log \frac{\exp(\text{sim}(z_i, \tilde{z}_i)/\tau)}{\sum_{u \in \mathcal{N}_{\text{same}}} \exp(\text{sim}(z_i, z_u)/\tau)} \right), \quad (6)$$

where $\mathcal{N}_{\text{same}}$ denotes the set of negative samples from the same domain and τ is the temperature parameter. This objective encourages nodes to remain similar to their local representatives while effectively distinguishing irrelevant ones, establishing robust local consistency constraints within each domain and laying the foundation for cross-domain alignment.

Cross-Domain View. Relying solely on the in-domain view, the encoder is limited to learning discriminative semantics within a

single domain, while domain discrepancies still remain. To align semantically similar nodes across domains into a unified space and enhance transferability and generalization, we propose the cross-domain view. For each domain, we perform subspace clustering to construct a compact set of domain-level prototypes that capture major semantic and structural patterns. Each node then aggregates its most similar cross-domain prototypes as alignment signals, reducing semantic gaps and enabling consistent cross-domain modeling.

To improve efficiency, we pre-construct a prototype library in the semantic space for each domain d : $\mathcal{P}_d = \{p_{d,k}\}_{k=1}^{K_d}$. Specifically, we stack the semantic representations of all nodes in domain d as $X_d \in \mathbb{R}^{n_d \times d_l}$. To enhance structural robustness and mitigate the influence of sparse connectivity on clustering boundaries, we first calculate a T -truncated approximation of the GNN-based node representations,

$$H_d = \sum_{t=0}^T (1 - \alpha) \alpha^t \tilde{A}^t X_d \quad (7)$$

which encodes both textual and structural semantics. To reduce redundancy and noise in the high-dimensional H_d , we apply subspace clustering with low-rank constraints for more compact and robust representations. Concretely, we construct a self-expressive matrix $S \in \mathbb{R}^{n \times n}$, requiring each node representation to be reconstructed as a linear combination of others:

$$\min_S \|H_d - SH_d\|_F^2 + \lambda \|S\|_*, \quad (8)$$

where the first term ensures reconstruction accuracy and the second term imposes a nuclear norm constraint $\|S\|_*$ to enforce low-rank structure. The optimal solution can be approximated as: $S = U^r U^{r\top}$ [22], where $U^r \in \mathbb{R}^{n \times \tau}$ contains the top- τ singular vectors. It follows that performing k -means clustering on the rows of U^r is equivalent to spectral clustering on S [30]. Hence, clustering in U^r effectively integrates semantic and structural information, suppresses noisy variance, and yields more separable clusters compared to clustering in the original semantic space. We then perform k -means clustering on U^r to obtain discriminative partitions. For the k -th cluster $C_{d,k}$ in domain d , the centroid is:

$$\mu_{d,k} = \frac{1}{|C_{d,k}|} \sum_{i \in C_{d,k}} U_i^r. \quad (9)$$

To smooth noise and boundary effects in the semantic space, we weight cluster members by their proximity to the centroid. The closeness of node i to cluster $C_{d,k}$ is defined as:

$$\hat{c}_{i,k} = \frac{c_{i,k}}{\sum_{j \in C_{d,k}} c_{j,k}}, \quad c_{i,k} = \frac{1}{1 + \|U_i^r - \mu_{d,k}\|_2} \quad (10)$$

Based on this, we perform weighted aggregation in the original feature space to obtain smoothed prototypes: $p_{d,k} = \sum_{i \in C_{d,k}} \hat{c}_{i,k} H_d \in \mathbb{R}^{d_l}$. Compared to centroid-only approaches, this proximity-weighted aggregation better utilizes intra-cluster information, yielding robust prototypes that preserve semantics while reflecting structural distributions. During training, for an anchor node i , we compute its similarities with cross-domain prototypes, select the top- M most similar ones to form the set $\mathcal{M}_i$, and obtain a prototype-based representation z_i^{proto} via soft assignment:

$$w_{im} = \frac{\exp(\beta \text{sim}(z_i, p_{i,m}))}{\sum_{m \in \mathcal{M}_i} \exp(\beta \text{sim}(z_i, p_{i,m}))}, \quad z_i^{\text{proto}} = \sum_{m \in \mathcal{M}_i} w_{im} p_{i,m}. \quad (11)$$

To achieve cross-domain alignment, we adopt a contrastive objective that pulls semantically similar nodes across domains closer while pushing apart unrelated ones:

$$\mathcal{L}_{\text{cross}} = -\frac{1}{B} \sum_{i=1}^B \left(\log \frac{\exp(\text{sim}(z_i, z_i^{\text{proto}})/\tau)}{\sum_{u \in \mathcal{N}_{-d(i)}} \exp(\text{sim}(z_i, z_u)/\tau)} \right), \quad (12)$$

where $\mathcal{N}_{-d(i)}$ denotes negative samples from other domains.

Joint Optimization. To simultaneously promote in-domain consistency and cross-domain alignment, we optimize the dual-view framework using the combined objective:

$$\mathcal{L} = \mathcal{L}_{\text{in}} + \lambda \mathcal{L}_{\text{cross}} \quad (13)$$

Minimizing this joint objective enables the model to learn both discriminative representations within individual domains and transferable features across domains, significantly improving generalization. Theoretical analysis can be found in **Appendix A**.

Efficiency Improvement. Contrastive learning typically requires repeated encoding of numerous samples, leading to high computational and memory overhead. To mitigate this, we maintain a dynamically updated embedding table E , which stores the most recent representation of each node. Each node is assigned a unique identifier $\text{Idx}(v)$ derived from its graph and node indices. During training, only embeddings of nodes in the current batch are updated via forward propagation, while representations of other candidates are retrieved directly from the cache. This approach substantially reduces redundant computation and improves training efficiency.

In the cross-domain view, we further address the computational cost of conventional k -means clustering by adopting an online update strategy based on exponential moving average (EMA). Global prototypes are progressively refined using batch-level centroids, avoiding frequent full-dataset clustering while maintaining alignment quality. Formally, each prototype is updated as:

$$p_{d,k} \leftarrow \text{norm}(\alpha_{\text{ema}} \cdot p_{d,k} + (1 - \alpha_{\text{ema}}) \cdot \hat{p}_{d,k}). \quad (14)$$

4.2 Graph Adaptation Training

Although the unified LLM encoder can generate embeddings for TAGs across domains after DCFT, directly applying these embeddings to graph-based tasks faces two major challenges: (1) the lack of explicit structural information, and (2) dimensional misalignment with GNN input requirements. To address these issues, we propose a Graph Adaptation Training strategy.

Embedding Adaptation. Previous studies [23] employ MLPs for dimensionality reduction of LLM embeddings. However, research have shown that this may lead to information loss [1]. To mitigate this problem, we introduce a two-step transformation approach. In the first step, we apply PCA to reduce embedding dimensions to an intermediate size (e.g., 1024 dimensions), preserving more semantic information than aggressive compression. This process is formalized as $e_i^{\text{LLM}} \xrightarrow{\text{PCA}} \hat{e}_i^{\text{LLM}}$. In the second step, we design a trainable adapter to generate GNN-compatible node embeddings:

$$e_i = W_2 \sigma(W_1 \hat{e}_i^{\text{LLM}}) + b, \quad (15)$$

This transformation produces dimension-reduced node embeddings that maintain semantic integrity while ensuring compatibility with GNN architectures.

Laplacian–Semantic Reconstruction. While the adapter generates computation-friendly low-dimensional representations, the resulting embeddings e_i are constructed independently of graph topology and may suffer from over-smoothing during GNN propagation. Additionally, dimensionality reduction inevitably causes some semantic information loss. To address these limitations, we introduce both structural Laplacian reconstruction and semantic reconstruction constraints.

(1) *Structural Laplacian Reconstruction.* To align compressed representations with the original graph structure, we minimize the following reconstruction loss:

$$\mathcal{L}_{\text{str}} = \frac{1}{|E_s|} \sum_{i,j} \tilde{A}_{ij} \|e_i - e_j\|_2^2 + \lambda \frac{1}{|N_s|} \sum_{i,k} [\delta - \|e_i - e_k\|_2^2]_+, \quad (16)$$

where $\tilde{A}$ is the normalized adjacency; E_s and N_s represent the sampled sets of edges and non-edge pairs, respectively; $[x]_+$ denotes the hinge function and δ controls the separation margin. The first term approximates the normalized Laplacian quadratic form $2 \text{Tr}(E^T L E)$ on sampled edges, encouraging proximity between connected nodes. The second term imposes a margin constraint on non-adjacent nodes, suppressing over-smoothing and preventing excessive contraction during propagation.

(2) *Semantic Reconstruction.* To compensate for fine-grained semantic loss during dimensionality reduction, we introduce a semantic reconstruction constraint. We require that the adapter-generated representation e_i can reconstruct a fixed target semantic representation e'_i through a lightweight decoder:

$$\mathcal{L}_{\text{sem}} = \frac{1}{N} \sum_i \|\hat{e}_i - e'_i\|^2, \quad \hat{e}_i = \text{Decoder}(e_i). \quad (17)$$

Here, e'_i represents the PCA-projected LLM embedding, which remains frozen during training. This constraint ensures semantic reversibility of compressed representations without introducing additional inference overhead. Combined with structural reconstruction, it guarantees that the final embeddings preserve both graph-structural signals and semantic recoverability.

GNN Propagation and Alignment. The adapted node representations e_i serve as input to the GNN, producing structure-aware representations h_i for downstream classification. However, since h_i is shaped by adjacency relationships and multi-hop aggregation, its distribution may diverge from the semantically-oriented embeddings e_i . To explicitly bridge this gap, we implement cross-space

alignment using mini-batch contrastive learning:

$$\mathcal{L}_{\text{align}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(h_i, e_i)/\tau)}{\sum_{k=1}^B \exp(\text{sim}(h_i, e_k)/\tau)} \quad (18)$$

This objective encourages consistency between GNN-generated structural representations h_i and semantic embeddings e_i , mitigating semantic drift during propagation and enhancing downstream task performance.

The overall training objective integrates all components:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \alpha \mathcal{L}_{\text{align}} + \beta \mathcal{L}_{\text{str}} + \gamma \mathcal{L}_{\text{sem}}. \quad (19)$$

During inference, only the adapter, GNN, and classification head are retained. The final node representations are directly utilized for classification, ensuring efficient and lightweight deployment.

5 Experiments

This section presents a comprehensive experimental evaluation of the proposed UTAG framework on multiple text-attributed graph datasets. We aim to answer the following research questions: **RQ1**: How does UTAG compare against existing LM-GNN cascade methods and LLM-enhanced approaches on downstream tasks? **RQ2**: How effectively does UTAG transfer across domains and within domains? **RQ3**: What is the contribution of each component in UTAG? **RQ4**: How do key hyperparameters affect model performance? **RQ5**: How efficient is UTAG in terms of training and inference?

5.1 Datasets

We evaluate UTAG on six benchmark TAG datasets spanning three distinct domains, including three from the academic domain: *Cora* [25], *Ogbn-Arxiv* [17] and *Arxiv2023* [14]; two from the e-commerce domain: *Ogbn-Products* [17] and *Ele-Computers* [34]; and one from the social domain: *Instagram* [19]. For the in-domain transfer task, we additionally use two datasets *Pubmed* [25] and *Ele-Photos* [34] for evaluation. Detailed descriptions provided in **Appendix B**.

5.2 Baselines

We compare UTAG against 16 competitive baselines, categorized into four groups:

- Traditional GNN models: **GCN** [20], **SAGE** [12], and **GAT** [29].
- Fully fine-tuned LMs: **BERT** [7] and **DeBERTa** [13].
- LM+GNN methods: **GIANT** [6], **GLEM** [40], **GraphBridge** [31], and **UniGLM** [9].
- LLM+GNN methods: **TAPE** [14], **GraphAdapter** [18], **OFA** [23], **ENGINE** [42], **GAugLLM** [10], and **LanSAGNN** [21].

For cross-domain transfer tasks, we additionally include two strong baselines: **UniGraph** [15] and **GraphCLIP** [41]. Further details about all these baselines are provided in **Appendix C**.

5.3 Experimental Settings

All methods are evaluated under consistent experimental settings. Each baseline reports its best performance using the optimal GNN variant selected from GCN, SAGE, and GAT. Our implementation of UTAG builds upon the 8B versions of LLaMA3 [28], with rank-8 LoRA fine-tuning for efficient adaptation. Most other LLM-enhanced baselines are also built on the same LLaMA3-8B backbone for a fair comparison. All results are reported as mean $\pm$ standard

Table 1: The impact of encoder backbones on UTAG.

Model	Ogbn-Arxiv	Ogbn-Products	Instagram
DeBERTa	75.77 $\pm$ 0.28	79.19 $\pm$ 0.76	64.83 $\pm$ 0.31
Llama2-13B	77.82 $\pm$ 1.25	81.65 $\pm$ 0.72	66.81 $\pm$ 1.22
Llama3-8B	78.14 $\pm$ 0.63	82.92 $\pm$ 0.81	66.57 $\pm$ 0.28

deviation over five independent runs (random seeds 42–46), conducted on a server with four NVIDIA RTX 4090 (24GB) GPUs. We use accuracy for node classification and AUC for link prediction. Complete implementation details are available in **Appendix D**.

5.4 Performance Analysis (RQ1)

We first pre-train the unified encoder on all six datasets and then employ the generated embeddings for graph adaptation training in node classification tasks. As shown in Table 2, UTAG achieves state-of-the-art performance on five out of six benchmarks. Traditional GNNs, relying solely on structural features, generally show limited accuracy due to their inability to capture rich semantic content. In contrast, pretrained language models overlook structural relationships, leading to subpar performance when compared to LM+GNN methods that fuse structural and semantic information. For example, UniGLM achieves an accuracy of 78.44% on the *Ogbn-Products* dataset, significantly outperforming both GNN-only and LM-only methods. LLM-enhanced methods further elevate performance by integrating advanced semantic representation with structural modeling. Nevertheless, UTAG surpasses even the strongest LLM-enhanced baselines, achieving 78.14% accuracy on the large-scale *Arxiv* dataset. Compared to other LLM-enhanced approaches, UTAG achieves an average improvement of 2.33% across all six datasets, with a notable 3.57% absolute gain on *Ogbn-Products*. These results validate the effectiveness of employing LLMs as unified embedding generators for TAG representation learning. It is worth noting that UTAG attains the second-best performance on the *Instagram* dataset, which we attribute to the dataset’s relatively small scale and high semantic noise in its text content, limiting the advantages of cross-domain alignment.

Table 1 presents the performance of UTAG with different encoder backbones. Both LLM backbones consistently outperform the LM baseline across all datasets, demonstrating the advantage of large language models in producing transferable embeddings for TAGs. LLaMA3-8B generally achieves better results than LLaMA2-13B, which can be attributed to architectural enhancements and more sophisticated training methodologies in the LLaMA3 series. On the *Instagram* dataset, LLaMA2-13B slightly exceeds LLaMA3-8B, possibly owing to the dataset’s relatively smaller scale.

To assess the impact of unified pretraining on embedding quality, we visualize embeddings from different models on two cross-domain datasets using t-SNE. As shown in Figure 4: (a) Embeddings from vanilla LLaMA3-8B show limited intra-class separability and overlapping cross-domain distributions; (b) Single-domain fine-tuning improves within-domain discrimination but generalizes poorly; (c) UTAG with unified pretraining achieves clear class separation within each domain while maintaining cross-domain

Table 2: Performance Comparison of Node Classification on Benchmark Datasets. Our method UTAG is implemented based on Llama3-8B. The best and second-best results are highlighted in bold and underline, respectively.

	Models	Cora	Ogbn-Arxiv	Arxiv2023	Ogbn-Products	Ele-Computers	Instagram
GNN	MLP	74.37 $\pm$ 0.45	68.66 $\pm$ 1.12	63.12 $\pm$ 0.78	68.84 $\pm$ 0.33	72.44 $\pm$ 1.16	58.23 $\pm$ 0.09
	GCN	79.87 $\pm$ 1.23	71.23 $\pm$ 0.45	67.56 $\pm$ 0.83	71.45 $\pm$ 0.17	75.67 $\pm$ 0.62	62.91 $\pm$ 0.45
	SAGE	78.12 $\pm$ 0.87	71.78 $\pm$ 0.91	76.34 $\pm$ 0.32	71.59 $\pm$ 0.54	76.66 $\pm$ 0.76	59.28 $\pm$ 1.52
	GAT	81.53 $\pm$ 0.31	72.41 $\pm$ 0.28	67.95 $\pm$ 0.65	71.07 $\pm$ 0.39	78.82 $\pm$ 0.98	63.36 $\pm$ 0.37
LM	Bert	81.59 $\pm$ 0.61	73.05 $\pm$ 0.27	77.73 $\pm$ 1.20	74.31 $\pm$ 0.84	73.97 $\pm$ 0.15	62.11 $\pm$ 1.34
	DeBERTa	79.52 $\pm$ 0.97	72.27 $\pm$ 0.41	77.80 $\pm$ 0.73	74.16 $\pm$ 1.08	75.45 $\pm$ 0.22	62.73 $\pm$ 0.67
LM+GNN	GIANT	85.09 $\pm$ 0.12	74.58 $\pm$ 1.05	76.37 $\pm$ 0.54	74.95 $\pm$ 0.98	79.26 $\pm$ 0.31	62.61 $\pm$ 0.21
	GLEM	87.69 $\pm$ 1.14	74.26 $\pm$ 0.73	74.88 $\pm$ 1.12	75.51 $\pm$ 0.47	78.22 $\pm$ 0.85	64.73 $\pm$ 0.67
	GraphBridge	88.84 $\pm$ 1.16	75.19 $\pm$ 0.38	78.55 $\pm$ 0.84	77.08 $\pm$ 0.19	79.39 $\pm$ 1.49	66.79 $\pm$ 0.53
	UniGLM	87.38 $\pm$ 1.56	75.62 $\pm$ 0.54	78.17 $\pm$ 0.21	78.44 $\pm$ 0.69	79.63 $\pm$ 0.34	65.05 $\pm$ 1.29
LLM+GNN	ENGINE (Llama3-8B)	89.04 $\pm$ 0.45	76.14 $\pm$ 0.36	79.45 $\pm$ 0.91	78.52 $\pm$ 0.54	78.85 $\pm$ 1.28	64.71 $\pm$ 0.78
	TAPE (GPT3.5)	89.76 $\pm$ 1.05	76.31 $\pm$ 0.88	80.48 $\pm$ 0.63	78.85 $\pm$ 0.10	79.27 $\pm$ 0.77	65.64 $\pm$ 0.73
	OFA (Llama3-8B)	88.19 $\pm$ 0.42	75.55 $\pm$ 0.42	78.93 $\pm$ 0.67	78.26 $\pm$ 0.25	<u>80.71 $\pm$ 0.59</u>	64.42 $\pm$ 1.37
	GraphAdapter (Llama3-8B)	87.93 $\pm$ 0.71	76.26 $\pm$ 0.46	78.59 $\pm$ 0.27	79.35 $\pm$ 1.21	78.73 $\pm$ 0.57	64.32 $\pm$ 0.84
	GAugLLM (Llama3-8B)	89.63 $\pm$ 0.18	75.08 $\pm$ 0.15	79.77 $\pm$ 0.74	<u>79.43 $\pm$ 0.38</u>	79.54 $\pm$ 0.92	66.89 $\pm$ 1.60
	LanSAGNN (Llama3-8B)	88.26 $\pm$ 0.77	<u>76.75 $\pm$ 0.05</u>	80.15 $\pm$ 1.30	78.60 $\pm$ 0.65	80.33 $\pm$ 0.44	65.78 $\pm$ 1.07
	UTAG (Llama3-8B)	91.87 $\pm$ 0.91	78.14 $\pm$ 0.63	82.68 $\pm$ 0.29	82.92 $\pm$ 0.81	82.65 $\pm$ 0.46	<u>66.57 $\pm$ 0.28</u>

Table 3: Comparative Evaluation of In-Domain Transferability Against Strongest Baselines.

	Models	Pubmed	Ele-Photos
In-domain	GAugLLM	84.86 $\pm$ 0.44	75.37 $\pm$ 0.69
	UniGLM	85.53 $\pm$ 0.29	76.76 $\pm$ 1.04
	OFA	86.37 $\pm$ 0.85	76.14 $\pm$ 0.64
	UTAG	87.68 $\pm$ 0.15	78.92 $\pm$ 0.27

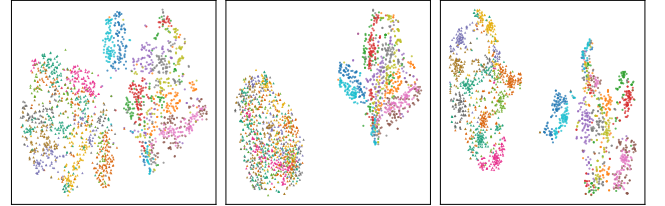
Table 4: Comparative Evaluation of Cross-Domain Transferability Against Strongest Baselines.

	Models	Ogbn-Arxiv	Ogbn-Products
Cross-domain (3-shot)	UniGLM	34.59 $\pm$ 1.06	50.84 $\pm$ 1.04
	GraphCLIP	34.84 $\pm$ 0.36	52.55 $\pm$ 0.97
	UniGraph	36.49 $\pm$ 0.73	51.21 $\pm$ 0.58
	UTAG	43.23 $\pm$ 0.62	57.44 $\pm$ 0.79
Cross-domain (5-shot)	UniGLM	35.45 $\pm$ 1.35	52.82 $\pm$ 1.12
	GraphCLIP	36.19 $\pm$ 0.63	53.71 $\pm$ 0.33
	UniGraph	38.63 $\pm$ 0.96	52.33 $\pm$ 0.12
	UTAG	45.96 $\pm$ 0.19	58.21 $\pm$ 0.71

distinctions, validating its effectiveness in cross-domain representation learning.

5.5 Transfer Analysis (RQ2)

We evaluate transfer capability from both in-domain and cross-domain perspectives. For in-domain transfer, we pre-train UTAG on three academic and two e-commerce datasets respectively, then evaluate on corresponding in-domain holdout sets (*Pubmed* and *Ele-Photos*). As shown in Table 3, UTAG outperforms all baselines,



(a) Llama3-8B (b) Fine-tuning a single TAG (c) DCFT

Figure 4: Visual Comparison of Embedding Methods Using t-SNE on Ogbn-ArXiv and Ogbn-Products.

demonstrating its ability to fully leverage both semantic and structural information within the same domain. For cross-domain transfer, we pre-train on academic datasets and evaluate on e-commerce datasets, and vice versa. Table 4 shows that UTAG consistently outperforms the strongest baselines in both 3-shot and 5-shot settings, with an average improvement of 5.92% in the 5-shot scenario. This indicates that UTAG’s cross-domain alignment mechanism effectively enhances generalization across heterogeneous domains.

5.6 Ablation Study (RQ3)

We systematically ablate key components of UTAG to assess their contributions. Results in Table 5 show that: 1) w/o Pretraining: Using the raw LLM without pretraining causes the most significant performance drop, highlighting the importance of cross-domain pre-training for learning universal representations; 2) w/o $\mathcal{L}_{in}/\mathcal{L}_{cross}$: Removing either contrastive loss degrades performance, confirming their complementary roles in ensuring intra-domain robustness and cross-domain generalization; 3) w/o Adapter: Directly using PCA-reduced embeddings with GNNs substantially hurts performance, demonstrating the necessity of non-linear adaptation; 4) w/o $\mathcal{L}_{str}/\mathcal{L}_{sem}$: Removing structural or semantic reconstruction

Table 5: Ablation Study on Component Contributions.

	Ogbn-Arxiv	Ogbn-Products	Computers
UTAG	78.14±0.63	82.92±0.81	82.65±0.46
-w/o pretraining	75.13±0.71	78.29±1.33	78.63±0.59
-w/o in-domain	75.94±0.48	78.75±0.77	80.24±0.92
-w/o cross-domain	76.22±0.55	79.23±0.81	80.11±1.41
-w/o adapter	76.46±0.17	78.84±0.27	80.79±1.18
-w/o str-recon	77.03±0.63	80.78±0.36	81.43±0.58
-w/o sem-recon	76.65±1.27	80.94±0.69	81.84±0.95
-w/o alignment	76.14±0.45	79.93±0.88	81.05±1.13

Table 6: Training and Inference Efficiency Analysis. Results were obtained using 4× NVIDIA GeForce RTX 4090 GPUs. “UTAG-” denotes the variant without the proposed efficiency improvement techniques.

Methods	OFA	GAugLLM	UniGLM	UTAG-	UTAG
Time_{Train}	22h 13m	17h 39m	14h 33m	29h 41m	12h 27m
Time_{Infer}	12m 25s	9m 16s	8m 51s	2m 34s	2m 34s

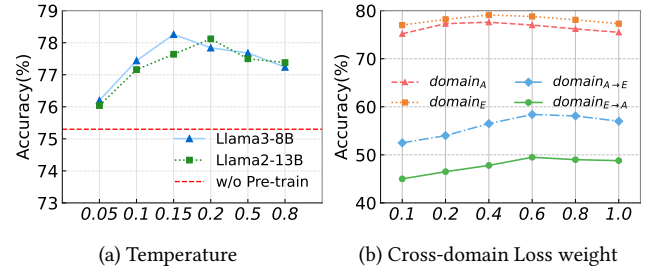
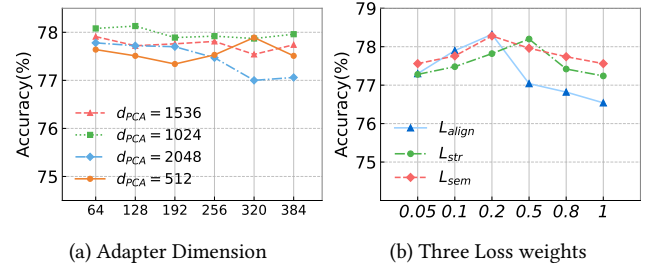
constraints causes performance decline, emphasizing the importance of preserving both structural and semantic information; 5) w/o $\mathcal{L}_{align}$: This ablation causes one of the largest accuracy drops, underscoring the critical role of aligning semantic and structural representations. Overall, each component contributes indispensably to UTAG’s optimal performance.

5.7 Parameter Analysis (RQ4)

We analyze the effects of key hyperparameters in both training stages of UTAG. Figure 5(a) shows the impact of contrastive temperature τ : performance first increases and then declines, peaking around 0.1–0.2, indicating that proper uniformity tuning is crucial for contrastive learning optimization. Figure 5(b) analyzes the cross-domain loss coefficient λ . For in-domain tasks, performance peaks at $\lambda \approx 0.3$ then declines, suggesting that excessive cross-domain consistency harms fine-grained semantic discrimination. For cross-domain tasks, performance improves with larger λ , peaking around 0.6, confirming that stronger cross-domain alignment enhances generalization across heterogeneous datasets. Figure 6(a) examines adapter output dimensions under different PCA-reduced intermediate dimensions. Accuracy improves with dimension size, peaks around 128, then slightly declines, showing UTAG’s robustness to adapter dimension selection within a reasonable range. Figure 6(b) shows the effect of auxiliary loss weights. Both overly small and large weights degrade performance, indicating that balanced regularization is essential for optimal adaptation.

5.8 Efficiency Analysis (RQ5)

In this section, we compare the efficiency of UTAG against several baselines in Table 6. UTAG demonstrates significant advantages in both training and inference efficiency. For OFA and UniGLM, training time refers to the total time required to train the model on all datasets. For UTAG, it additionally includes the time spent on the

**Figure 5: Hyperparameter experiments on Stage 1: impact of contrastive temperature τ , the weight of cross-domain loss.****Figure 6: Hyperparameter experiments on Stage 2: impact of the dimension of adapter output and three loss weights on Ogbn-arxiv dataset.**

second-stage adapter training on *Ogbn-Arxiv* beyond the pretraining phase. For GAugLLM, it is sum of the individual training times over six datasets. For training, UTAG requires approximately 44% less time than OFA and 29% less than GAugLLM, mainly due to our two efficiency improvement techniques: the dynamic embedding table and the online prototype update strategy. For inference on *Ogbn-Arxiv*, UTAG achieves the fastest speed (2 minutes 34 seconds), outperforming all baselines and achieving nearly 5× speedup compared to OFA. These results demonstrate that UTAG substantially reduces computational overhead while maintaining performance advantages, showing strong practicality and scalability.

6 Conclusion

This paper proposes UTAG, a unified framework that leverages Large Language Models as embedding generators for text-attributed graphs. To address the limitations of existing methods in cross-domain generalization and semantic-structural space alignment, we introduce a two-stage approach: first, a dual-view contrastive fine-tuning strategy aligns node texts from multiple domains into a shared semantic space; second, a lightweight graph adaptation mechanism with two-stage dimensionality reduction bridges the gap between high-dimensional LLM embeddings and low-dimensional graph spaces. Extensive experiments demonstrate that UTAG achieves state-of-the-art performance while exhibiting strong cross-domain generalization capability. Our work validates the effectiveness of LLMs as unified embedding generators, providing a scalable and transferable solution for TAG representation learning and opening up new possibilities for developing graph foundation models.

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A Theoretical Analysis

While UTAG empirically exhibits strong cross-domain generalization, we provide a theoretical justification grounded in domain adaptation theory. We formalize how the proposed cross-domain contrastive view reduces the source–target discrepancy in prototype space, tightening the upper bound of target-domain risk.

Preliminaries. Let P_S and P_T denote the source and target data distributions over the input space $\mathcal{X}$, and let $h \in \mathcal{H}$ be a hypothesis (classifier) operating on representations $z = f_\theta(x)$. Following Ben-David et al. [2], the expected target error $\epsilon_T(h)$ satisfies

$$\epsilon_T(h) \leq \epsilon_S(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T) + \lambda, \quad (20)$$

where $\epsilon_S(h)$ is the source risk, $\lambda = \min_{h \in \mathcal{H}} (\epsilon_S(h) + \epsilon_T(h))$ is the error of the ideal joint hypothesis, and $d_{\mathcal{H}\Delta\mathcal{H}}$ denotes the $\mathcal{H}$ -divergence measuring the discrepancy between domains.

Reducing the divergence term is thus essential for cross-domain generalization.

Projection to the Prototype Space. Each embedding $z_x = f_\theta(x)$ is softly assigned to a set of prototypes $\{p_k\}_{k=1}^K$:

$$q(x)_k = \frac{\exp(\beta \cos(z_x, p_k))}{\sum_{j=1}^K \exp(\beta \cos(z_x, p_j))}. \quad (21)$$

This induces the prototype-space distributions $Q_S = \{q(x) \mid x \sim P_S\}$, $Q_T = \{q(x) \mid x \sim P_T\}$. When the classifier operates on $q(x)$ rather than z_x , the discrepancy satisfies

$$d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T) = d_{\mathcal{H}\Delta\mathcal{H}}(Q_S, Q_T), \quad (22)$$

because the mapping $x \mapsto q(x)$ is deterministic.

Bounding by Statistical Divergences. The $\mathcal{H}\Delta\mathcal{H}$ divergence can be upper bounded by statistical divergences between Q_S and Q_T .

LEMMA A.1 (DIVERGENCE BOUND IN PROTOTYPE SPACE). *If classifiers in $\mathcal{H}$ are 1-Lipschitz w.r.t. the ℓ_2 norm on $q(x)$, then*

$$d_{\mathcal{H}\Delta\mathcal{H}}(Q_S, Q_T) \leq 2 \text{TV}(Q_S, Q_T) \leq \sqrt{2 \text{KL}(Q_S \parallel Q_T)}.$$

This implies that minimizing the KL divergence between prototype distributions directly reduces the domain discrepancy in the generalization bound (20).

Effect of the Cross-domain Contrastive Loss. Our cross-domain contrastive objective,

$$\mathcal{L}_{\text{cross}} = -\mathbb{E}_{x \sim P_S} \log \frac{\exp(\cos(z_x, p_T^+)/\tau)}{\sum_{p \in \mathcal{P}_T} \exp(\cos(z_x, p)/\tau)}, \quad (23)$$

encourages source samples to align with semantically consistent prototypes p_T^+ in the target domain. Under mild smoothness assumptions, optimizing $\mathcal{L}_{\text{cross}}$ decreases the class-conditional divergence.

THEOREM A.2 (CROSS-DOMAIN ALIGNMENT VIA CONTRASTIVE LEARNING). *Assume embeddings $z = f_\theta(x)$ are bounded and the cosine similarity is τ -smooth. Minimizing $\mathcal{L}_{\text{cross}}$ reduces the conditional divergence between class-conditional prototype distributions:*

$$\text{KL}(Q_S^{(y)} \parallel Q_T^{(y)}) \downarrow, \quad \forall y \in \mathcal{Y}.$$

Consequently, by Lemma A.1,

$$d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T) \downarrow, \quad \text{tightening the target risk bound in (20).}$$

Proof Sketch. The InfoNCE objective (23) is equivalent to minimizing the KL divergence between positive pairs' joint distribution and the product of their marginals [?]. Thus, minimizing $\mathcal{L}_{\text{cross}}$ aligns class-conditional prototype distributions, decreasing $\text{KL}(Q_S^{(y)} \parallel Q_T^{(y)})$ and hence $d_{\mathcal{H}\Delta\mathcal{H}}(Q_S, Q_T)$ by Lemma A.1. $\square$

Conclusion. Combining (20)–(23) and Theorem A.2, we obtain

$$\epsilon_T(h) \leq \epsilon_S(h) + \frac{1}{2} \sqrt{2 \text{KL}(Q_S \parallel Q_T)} + \lambda,$$

where $\text{KL}(Q_S \parallel Q_T)$ decreases as $\mathcal{L}_{\text{cross}}$ is optimized. Therefore, the proposed cross-domain view explicitly reduces domain discrepancy in the prototype space, tightening the target-domain risk bound and explaining UTAG's superior cross-domain generalization.

B Datasets

For the node classification task, we use six datasets. The details of the datasets are provided below:

- **Cora** [25] dataset consists of 2,708 scientific publications categorized into seven classes: Case-based, Genetic Algorithms, Neural Networks, Probabilistic Methods, Reinforcement Learning, Rule Learning, and Theory. Each paper in the citation network cites or is cited by at least one paper, resulting in a total of 5,429 edges.
- **Ogbn-ArXiv** [17] dataset represents a directed graph showcasing the citation network among computer science arXiv papers. Each node signifies an arXiv paper, with directed edges denoting citations. The goal is to categorize papers into one of 40 subject areas such as cs.AI, cs.LG, and cs.OS.
- **Arxiv2023** [14] dataset is a directed graph representing the citation network of computer science arXiv papers published in 2023 or later. Similar to OGBN-ArXiv, it features nodes representing arXiv papers and directed edges for citations. The goal is to classify each paper into one of 40 subject areas.
- **Ogbn-Products** [17] dataset includes 54k nodes and 74k edges. Each node in this dataset represents products sold on Amazon, and edges between two products indicate that the products are purchased together. The task involves predicting the category of a product in a multi-class classification setup.
- **Ele-Computers** [34] dataset, extracted from the Amazon Electronics dataset, consists of nodes representing electronic products, with edges indicating frequent co-purchases or co-views. Each node is similarly labeled according to a three-level classification of electronics products. The text attribute for each node is the user review with the most votes.
- **Instagram** [19] dataset reflects a social network where edges represent following relationships, nodes signify users, and the prediction task involves classifying users.

Method	Cora	Arxiv2023	Ele-Computers
GCN	82.45±0.73	91.78±0.22	87.23±0.55
SAGE	85.61±1.32	93.34±0.59	91.97±0.41
GAT	86.12±0.42	93.55±0.33	90.86±0.57
GraphBridge	86.73±0.88	93.28±0.67	92.44±0.44
UniGLM	88.09±1.12	93.67±0.60	91.31±0.12
OFA	87.52±0.95	92.19±0.15	92.85±0.39
ENGINE	88.38±1.49	94.29±0.23	92.70±0.81
UTAG	90.63±0.71	96.85±0.56	95.79±0.93

Table 7: Experimental results of link prediction task. The performance is evaluated by AUC metric.

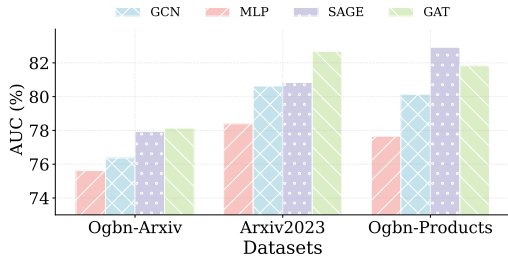


Figure 7: Comparison of GNN performance on three datasets.

C Baselines

In this section, we provide a detailed introduction to the baselines that we use to evaluate UTAG.

Traditional GNNs: For node classification, we adopted three simple yet widely used GNN models: GCN, SAGE, and GAT.

Fintuned LMs: We adopt three commonly used pre-trained language models in our study: BERT and RoBERTa.

GIANT [6] is a cascading method with two stages: pretraining LMs and training GNNs. First, it enhances node representations by integrating structural information into LM pretraining. Then fine-tuned LM-generated node features serve as initial features for GNN training.

GLEM [40] is an effective framework that fuses LMs and GNNs in the training phase through a variational EM framework.

GraphBridge [31] is a integration framework that bridges local and global perspectives by leveraging contextual textual information, enhancing fine-grained understanding of TAGs.

TAPE [14] utilizes Large Language Models like ChatGPT to generate pseudo labels and explanations for textual nodes, which are then used to fine-tune Pre-trained Language Models alongside the original texts.

ENGINE [42] is an efficient fine-tuning and inference framework for text-attributed graphs. It co-trains large language models and GNNs using a ladder-side approach, optimizing both memory and time efficiency.

GraphAdapter [18] uses GNN as an efficient adapter in collaboration with LLMs to tackle TAGs. The entire framework is

trained using auto-regression on node text. Once trained, it can be seamlessly fine-tuned with task-specific prompts.

OFA [23] represents all nodes and edges with human-readable texts, encoding them across various domains into a unified space via LLMs. This framework adapts to diverse tasks by incorporating task-specific prompting substructures into the input graph.

GAugLLM [10] is a framework for augmenting TAGs. It leverages LLMs to enhance self-supervised graph learning by using a mixture-of-prompt-expert technique to generate augmented features.

LanSAGNN [21] is a framework that leverages LLMs to extract tailor-made semantic information for node pairs, effectively capturing the unique interactions within node relationships.

UniGLM [9] is a novel foundation model pre-trained over multiple TAGs from a variety of domains. It fine-tunes well-established LMs using a domain-aware contrastive learning objective that unifies structure heterogeneity across various domains.

D Link Prediction

Our method is not limited to node classification tasks, it can also be applied to edge-level or graph-level tasks. In this section, we conduct experiments on link prediction tasks. We split existing edges into train:val:test=0.85:0.05:0.1 for all datasets. The reported results include the mean AUC with standard deviation across 5 different random seeds. We use traditional GNN methods, LM+GNN methods and LLM enhanced methods as baselines and present the results of our method. According to Table 7, UTAG outperforms all baselines by a considerable margin on the three datasets. In particular, UTAG exhibits more than a 3% absolute improvement on the Ele-Computers dataset. These statistics underscore the effectiveness of our method in edge-level tasks.

E GNN Analysis

To explore the impact of different GNN architectures, we conduct experiments on three datasets across various downstream tasks. Figure 7 shows that MLP, which lacks structural awareness, performs the worst, confirming the importance of structural signals. GCN and SAGE show stable and strong performance overall. GAT, due to its adaptive neighbor weighting mechanism, achieves the highest accuracy in node classification tasks and is well-suited for large and dense graphs such as Ogbn-Arxiv and Ogbn-Products datasets. In summary, GCN and SAGE are robust across tasks, while GAT offers advantages in scenarios with complex structures or classification objectives, validating the impact of GNN choice on downstream performance.

F Limitations

In this work, our method UTAG is designed for static graphs. As a result, it does not naturally extend to dynamic or temporal graphs, where nodes/edges and their associated texts may evolve over time. Supporting dynamic graphs would require additional mechanisms for incremental embedding updates, temporal neighborhood sampling, and prototype maintenance under distribution drift. We will expand our framework in the future work.